CURRICULUM VITAE

Manu Nicholas Jacob

Hardware Engineer, Dell Technologies · Austin, Texas

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SUMMARY

Hardware engineer working on enterprise AI server platforms: validation, system-level debugging, and root-cause engineering. Independently runs a measurement-driven research program on edge machine-learning systems, with 13 papers across memory bandwidth, energy, thermal behaviour and quantization, 11 of them currently under review, and 10 public DOI-archived artifacts released alongside them. Named inventor on an invention disclosure authorized for filing.

EXPERIENCE

Hardware Engineer, Root Cause Engineering · Dell Technologies

JUN 2025 TO PRESENT

Server-scale system debug across PCIe, thermal behaviour and GPU/accelerator platforms on enterprise PowerEdge systems. Fast-cycle triage, reliability assessment, and recurring failure-pattern identification for long-term stability improvements.

Root Cause Engineering Intern · Dell Technologies

SUMMER 2024

Developed a unified diagnostic tool for NVIDIA AI platforms: PCIe replay and bus-reset detection, optimised SBR procedures, and a real-time replay-detection workflow improving fault isolation and technician usability.

Controls Engineering Intern · Schneider Electric

SUMMER 2023

Designed and deployed a palletizer-control HMI on the factory floor, optimised PLC-HMI communication, and led a standardisation-specification overhaul against ANSI/RIA and OSHA requirements.

EDUCATION

B.S. Electrical Engineering · B.S. Computer Engineering

2021 TO 2025

University of Massachusetts Amherst. Minors in Engineering Management and Economics.

PAPERS · 13

UNDER REVIEW · 11 Submitted, currently with reviewers.

[1] CPU Utilization as a Software-Only Thermal Proxy
IEEE Embedded Systems Letters · 2026 · doi:10.5281/zenodo.21844859

[2] The Memory Wall Governs Edge DNN Inference
ACM Transactions on Embedded Computing Systems · 2026

[3] Latency-Optimal Is Not Energy-Optimal
ACM Transactions on Embedded Computing Systems · 2026

[4] When Thermal-Margin Control Helps and When It Hurts
IEEE Embedded Systems Letters · 2026 · doi:10.5281/zenodo.21844861

[5] The Cold-Start Tax
IEEE Internet of Things Journal · 2026 · doi:10.5281/zenodo.21844857

[6] The Memory Wall at the Edge of Language
IEEE Transactions on Computers · 2026 · doi:10.5281/zenodo.21844855

[7] The Hybrid-Core Decode Cliff
IEEE Computer Architecture Letters · 2026

[8] Four Walls Before Hello: Deploying LLMs at the Edge
IEEE Design & Test · 2026

[9] The Cold-Start Tax at the Edge of Language
ACM Transactions on Embedded Computing Systems · 2026

[10] When Does INT8 Actually Pay on an Edge CPU?
IEEE Embedded Systems Letters · 2026

[11] The Break-Even Parallel Speedup
IEEE Computer Architecture Letters · 2026 · doi:10.5281/zenodo.21987261

IN PREPARATION · 2 Complete and submission-ready, not yet submitted.

[1] The Narrow Regime of Thermal-Aware Anytime Inference
IEEE Embedded Systems Letters · 2027

[2] Profiling Inference on a Consumer Laptop GPU
EuroMLSys (EuroSys workshop) · 2027

PATENTS & ARTIFACTS

Invention disclosure (named inventor): approved and authorized for filing through Dell Technologies. Formal USPTO filing pending; no application number issued yet.

10 open research artifacts: public code and data for the papers above, archived on Zenodo: rpi5-quantization-benchmark, edge-llm-memory-wall, edge-cold-start-tax, pi5-thermal-proxy, edge-thermal-margin-control, latency-elastic-edge-inference, edge-format-tax, edge-breakeven-speedup, llama-roofline, ml-systems-lab.

CAPSTONE

Pothole Detection System · ECE senior design, UMass Amherst — SEP 2024 TO MAY 2025

Real-time pothole detection: stereo cameras and ultrasonic sensors feeding CNN-based detection on Raspberry Pi hardware, a custom PCB for power and sensor interfacing, and an in-vehicle driver alert display. 80 to 90% detection accuracy in prototype testing at 35 to 45 mph.

COMPETITION AWARDS

Best Embedded System · WorldWide Rover — HACKUMASS XII

Best Hardware Hack + Best Circuit Hack · 4Sight — HACKUMASS X

PEER REVIEW AND ARTIFACT EVALUATION

Artifact Evaluation Committee · USENIX ATC 2026 — 2026

Invited by the committee chair, serving as an independent researcher. The work is assessing submitted research artifacts for availability, functionality and reproducibility, in a review window running 22 September to 14 October 2026. The roster is not published yet, so there is no page worth linking here until it is.

Reviewer, review delivered · JOSS: Optiland — 2026

An open-source optical design package. All 31 checklist items worked and the recommendation sent to the editor on 19 August 2026. Three issues went to the authors and all three were resolved: a benchmark script behind a published table that was not in the repository, a paper example that could not run as a script, and a throughput collapse on a 4 GB card that shows up well before the memory actually runs out. The last of those came from building the package and running its own GPU benchmark rather than reading the number in the paper, and the maintainer reproduced it on a different card. The recommendation is not the decision; the paper sits with its editor.

Reviewer, in progress · JOSS: nsEVDx — 2026

A Python library for non-stationary extreme value distributions, assigned 21 August 2026. Most of the checklist work here is whether the samplers do what the paper says they do.

Reviewer, in progress · JOSS: PySlyde — 2026

A toolkit for preprocessing digital pathology whole-slide images, review opened 24 August 2026. One of three reviewers on it. The interesting part of this one is that the claims are about throughput and memory on slides too large to hold at once, which is the same question as everything else here in a different domain.

Reviewer, assigned at pre-review · JOSS: Battflow — 2026

An automated workflow for predicting battery properties. Sole listed reviewer, assigned at the pre-review stage, which means the review itself has not opened yet. Listed here at exactly that stage and no further.

Listed reviewer · JOSS reviewer database — 2026

In the database since 17 August 2026, listed under performance engineering, benchmarking and embedded systems. Four assignments have come through it, and the four entries above are all of them.

SERVICE

- **Judge** · HackTX, UT Austin — 2025
- **Judge and Judge Advisor** · VEX Robotics — ONGOING
- **Mentor and judge** · University hackathons — ONGOING
- **Talk, submitted** · Embedded Vision Summit 2027 — 2027
- **Teaching maths and science** · Lovedale Foundation
- **Training adults with learning disabilities** · Diya Foundation
- **Video production** · Yuvalok Foundation

CAMPUS LEADERSHIP

- **Resident Assistant** · UMass Amherst, community of 450 residents — JAN 2023 TO MAY 2025
- **Campus Ambassador** · Dell Technologies at UMass Amherst — FALL 2024
- **Membership Coordinator** · IEEE, UMass Amherst — FALL 2023

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional — ORACLE · 2025
- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate — ORACLE · OCT 2025
- Oracle Cloud Infrastructure 2025 Certified Foundations Associate — ORACLE · OCT 2025

COURSEWORK

- AI Infrastructure and Operations Fundamentals — NVIDIA · AUG 2026
- Data Science: Probability (PH125.3x) — HARVARDX · JUN 2020
- AI for Everyone: Master the Basics — IBM · MAY 2020

SKILLS

HARDWARE

Enterprise servers, PCIe, GPU subsystems, embedded hardware, microcontrollers, power systems.

SOFTWARE

Python, C/C++, ONNX Runtime, PyTorch, Linux perf, hardware performance counters, validation automation.

ENGINEERING

Root-cause analysis, failure analysis, benchmarking, reliability engineering, experiment design, research and patent writing.